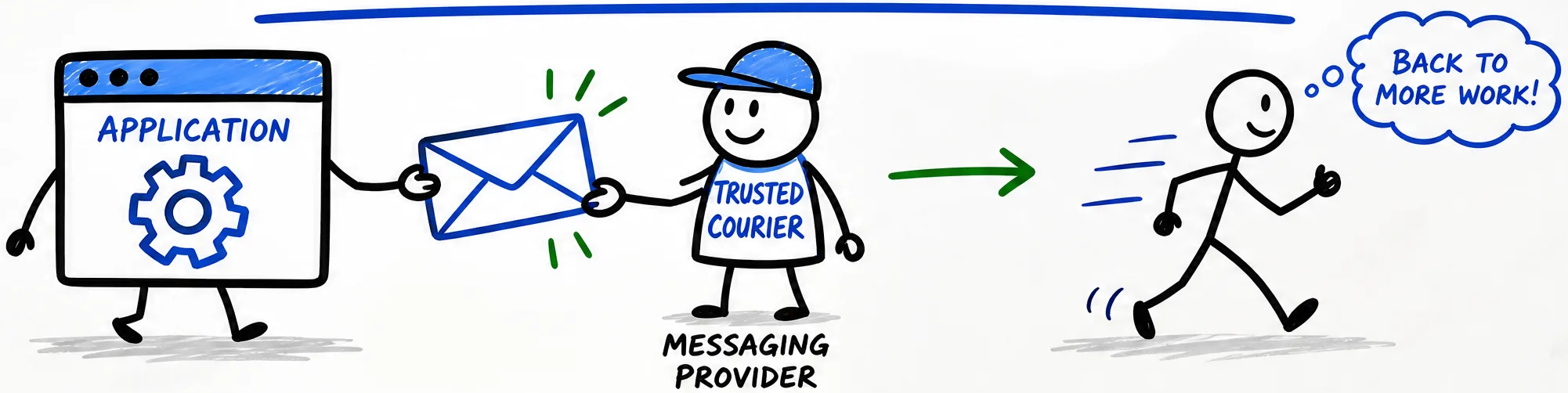
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WebSphere v9 Administration: Module 8

Connect to a Messaging Provider



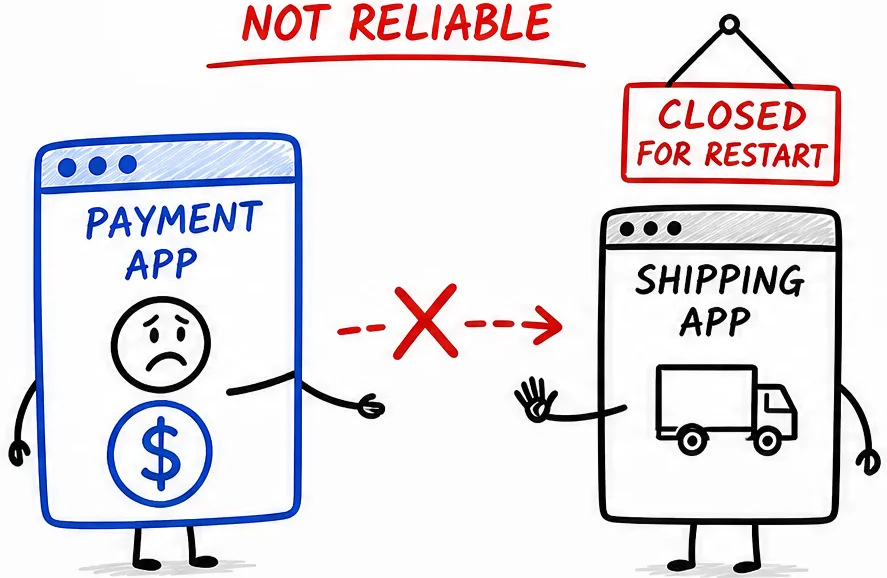
Decoupling time is how distributed systems stay responsive under pressure.

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Why Asynchronous Messaging Exists

NOT RELIABLE



Direct calls can fail when the service is unavailable.

TRADING IMMEDIATE ANSWERS — FOR — GUARANTEED PROGRESS

RELIABLE



The request is safely placed in a destination for guaranteed delivery.

The destination is the handoff point that turns dependency into resilience.

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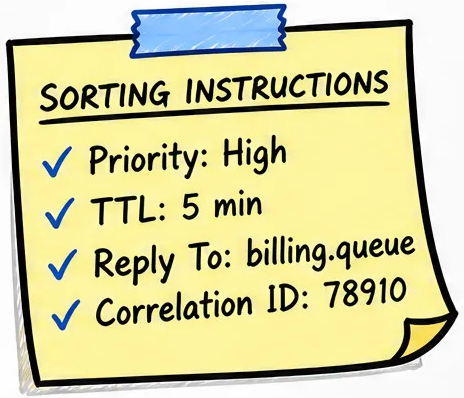
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The Anatomy of a JMS Message

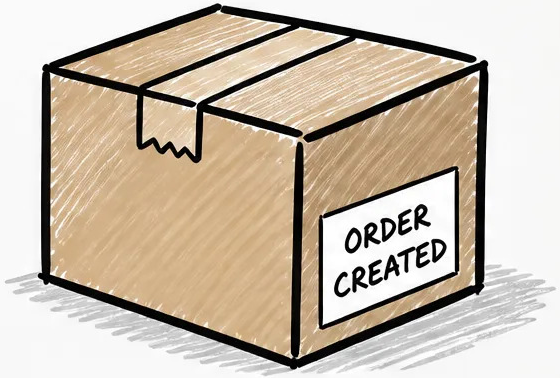
Header, Properties, Body



+



+



Header

Routing & protocol information

Properties

Context and instructions about the message

Body / Payload

The actual business message content



The payload says what happened;
metadata tells the platform what to do with it.

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Choosing a Messaging Provider

Back into WebSphere
(Default SIB)



An Architectural Boundary

Out to external enterprise
(IBM MQ)



Third-party
(any provider)



The provider choice is an architectural boundary, **NOT** a console preference.

The Service Integration Bus Architecture

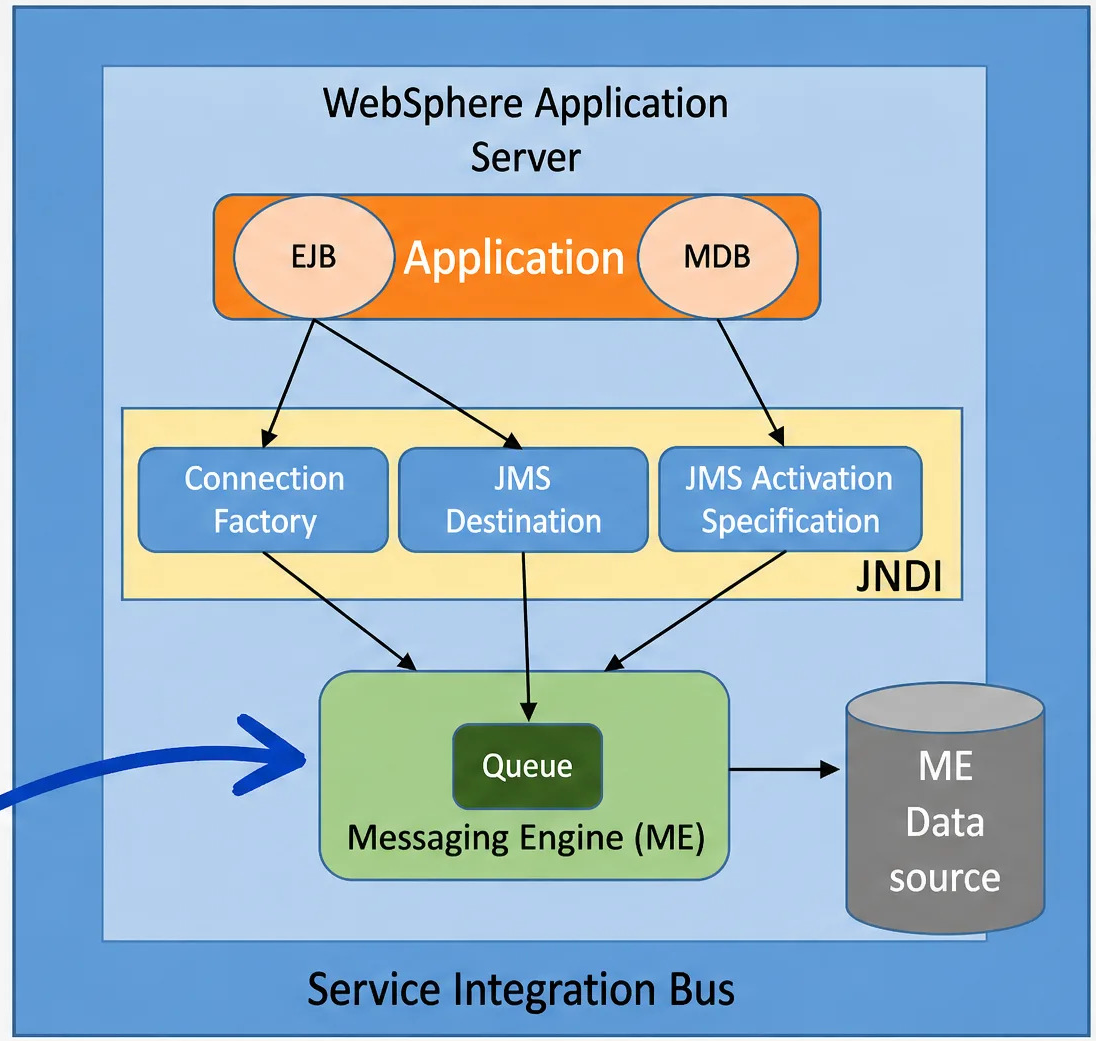
Connecting Systems. Routing Messages. Delivering Value.



The Physical Routing Engine

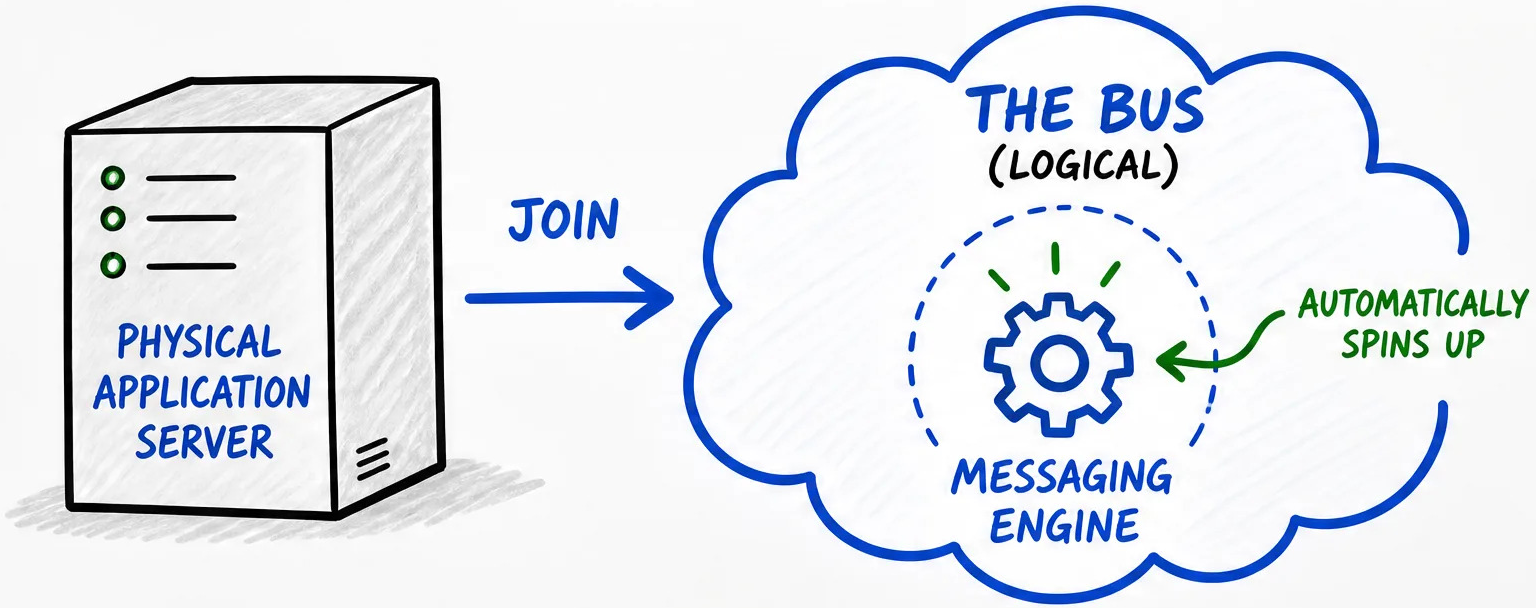


Connection factories create the route; destinations define the address; messaging engines carry the load.



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Bus Members and Messaging Engines



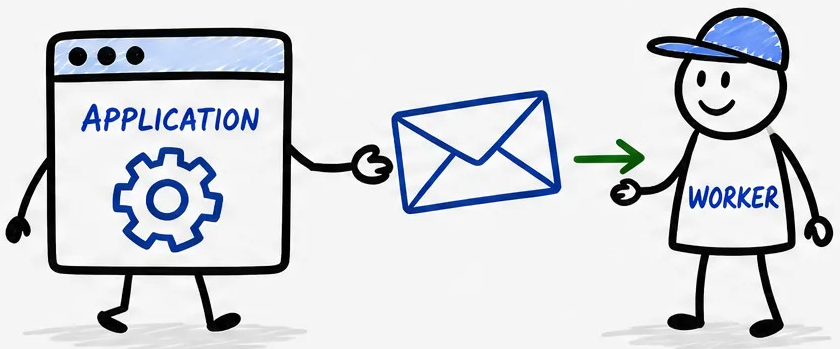
Providing the Runtime Home

A **messaging engine** is not merely configuration; it **owns live operational state.**

Queues, Topics, and Destinations

Point-to-point Queue

One message.
Exactly one consumer.



Distributes Work
to one recipient

Distributing Work vs Awareness

Publish/Subscribe Topic

One message.
Many subscribers.



Broadcasts Awareness
to Many Listeners

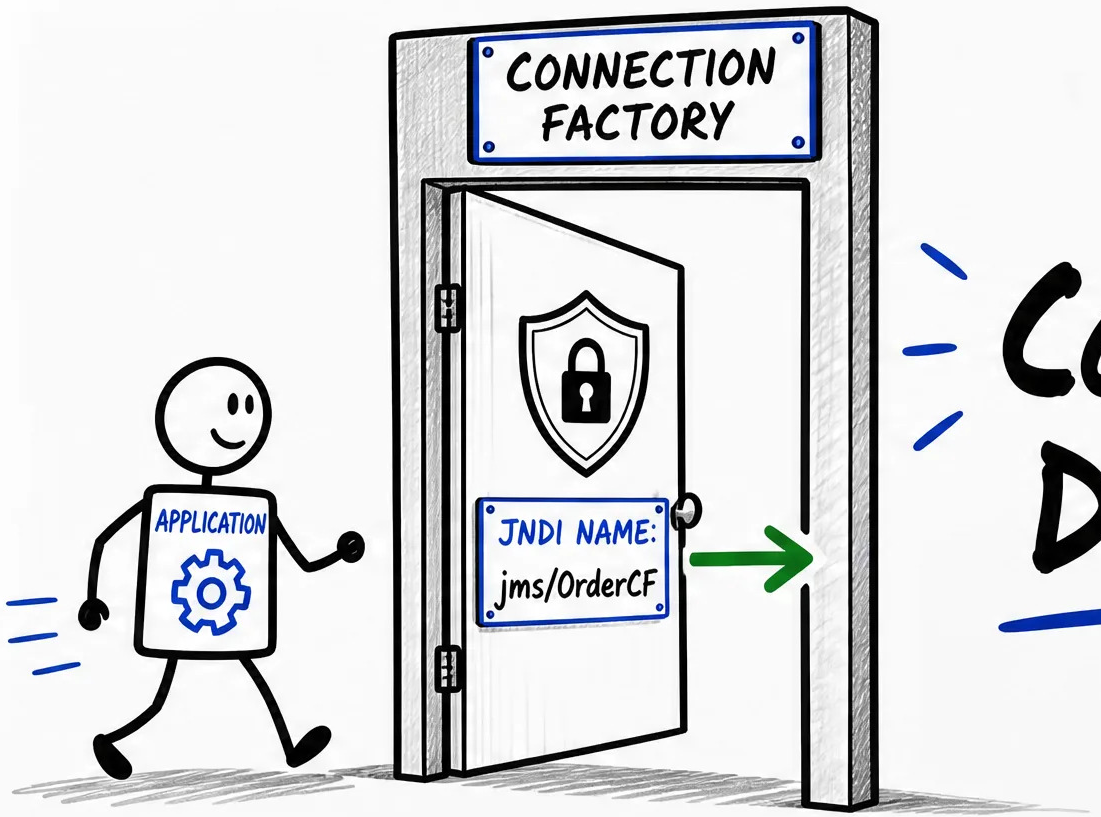


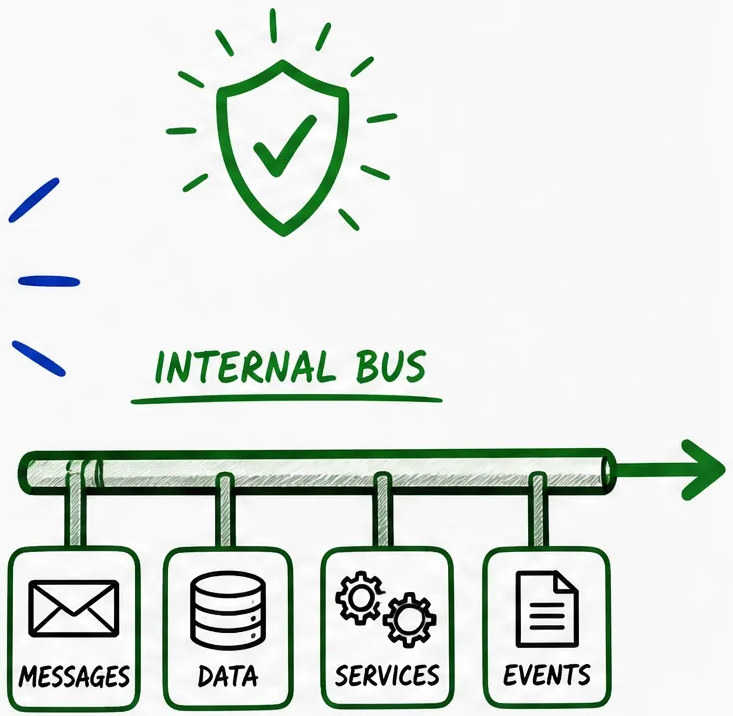
JNDI is the stable contract between application code and messaging topology.

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Connection Factories and JNDI

The Controlled Doorway

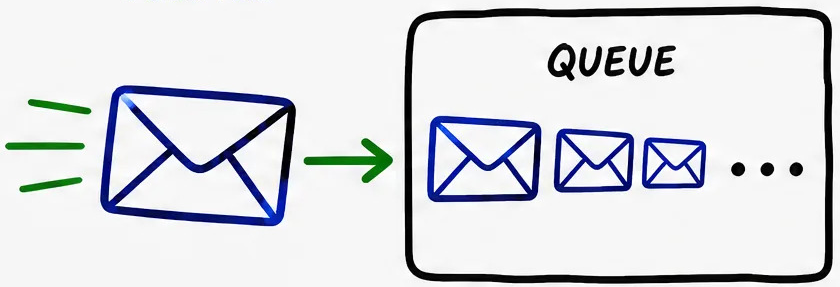




One connection factory should express one intentional integration path.

Activation Specifications and MDBs

1. Message Arrives



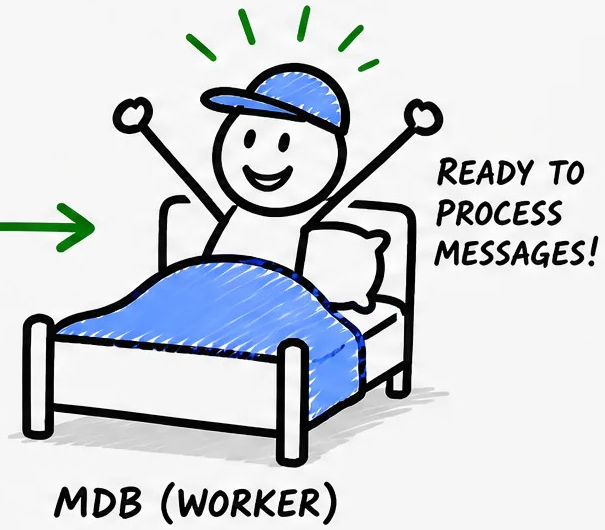
Container-Managed Triggers

2. Activation Specification triggers instantly



Precise alarm trigger

3. MDB (worker) wakes up!



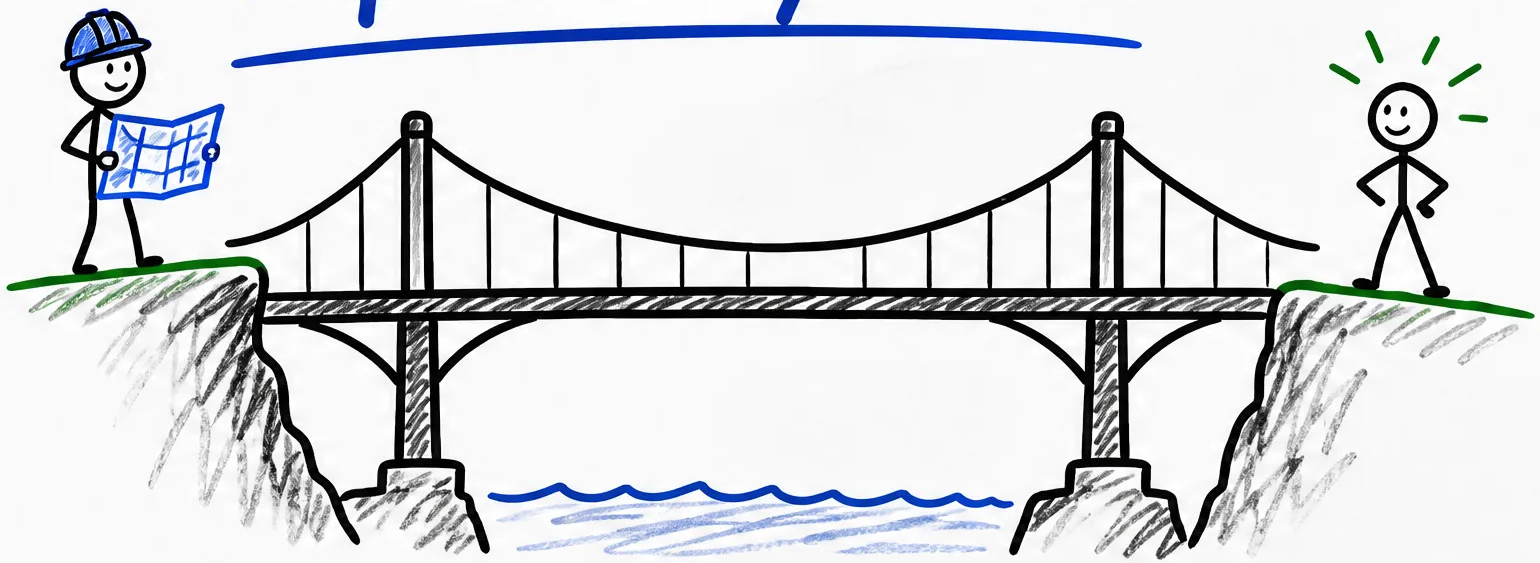


An activation specification turns an idle queue into a responsive work trigger.

The Configuration Playbook

1. Bus
2. Member/ME
3. Destination
4. Connection Factory
5. Activation Spec.

Build in Dependency Order



Build the reliable runtime path entirely before you point applications at it.

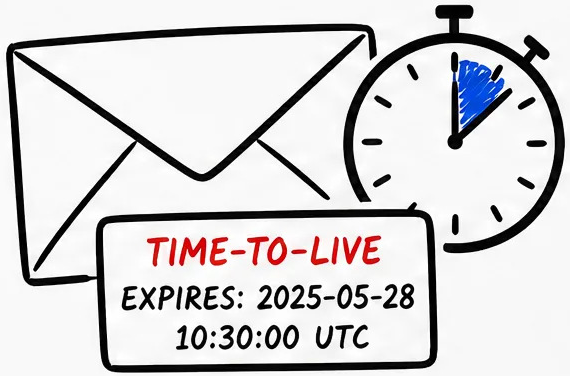
Reliability, Delivery, and Verification

1. Persistent message



Message is stored *persistently* to survive failures and restarts.

2. Strict time-to-live



A *strict expiration* ensures timely relevance and cleanup.

3. End-to-end delivery



Delivered *successfully* from sender to receiver.

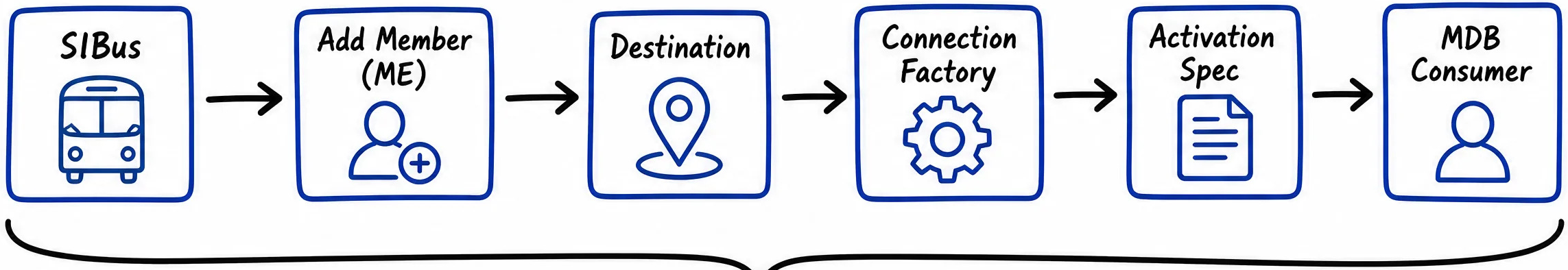
Policy-Driven Resilience

Reliability is a serious policy choice expressed clearly through concrete message settings.

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Module 8 Summary and Lab



The End-to-End Route

A reliable message path has a sender, a destination, a runtime engine, and a securely verified consumer.